COGNIFYZ DATA ENGINEERING INTERNSHIP

Railway Data Analysis Report

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1. DATASET OVERVIEW

Dataset: Railway_info.csv

Total Records: 11,113 trains

Columns: Train_No, Train_Name, Source_Station, Destination_Station, Days

Missing Values: None

Unique Source Stations: 921

Unique Destination Stations: 924

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2. KEY FINDINGS

→ CST-Mumbai is the busiest station with 513 trains originating daily

→ Friday has highest trains (1,649)

→ Thursday has lowest trains (1,526)

→ 71.2% trains operate on weekdays

→ 28.8% trains operate on weekends

→ Nagpur connects to 22 major cities

→ Moderate correlation (0.381) between
day of week and train frequency

→ Most stations handle less than
50 trains (right-skewed distribution)

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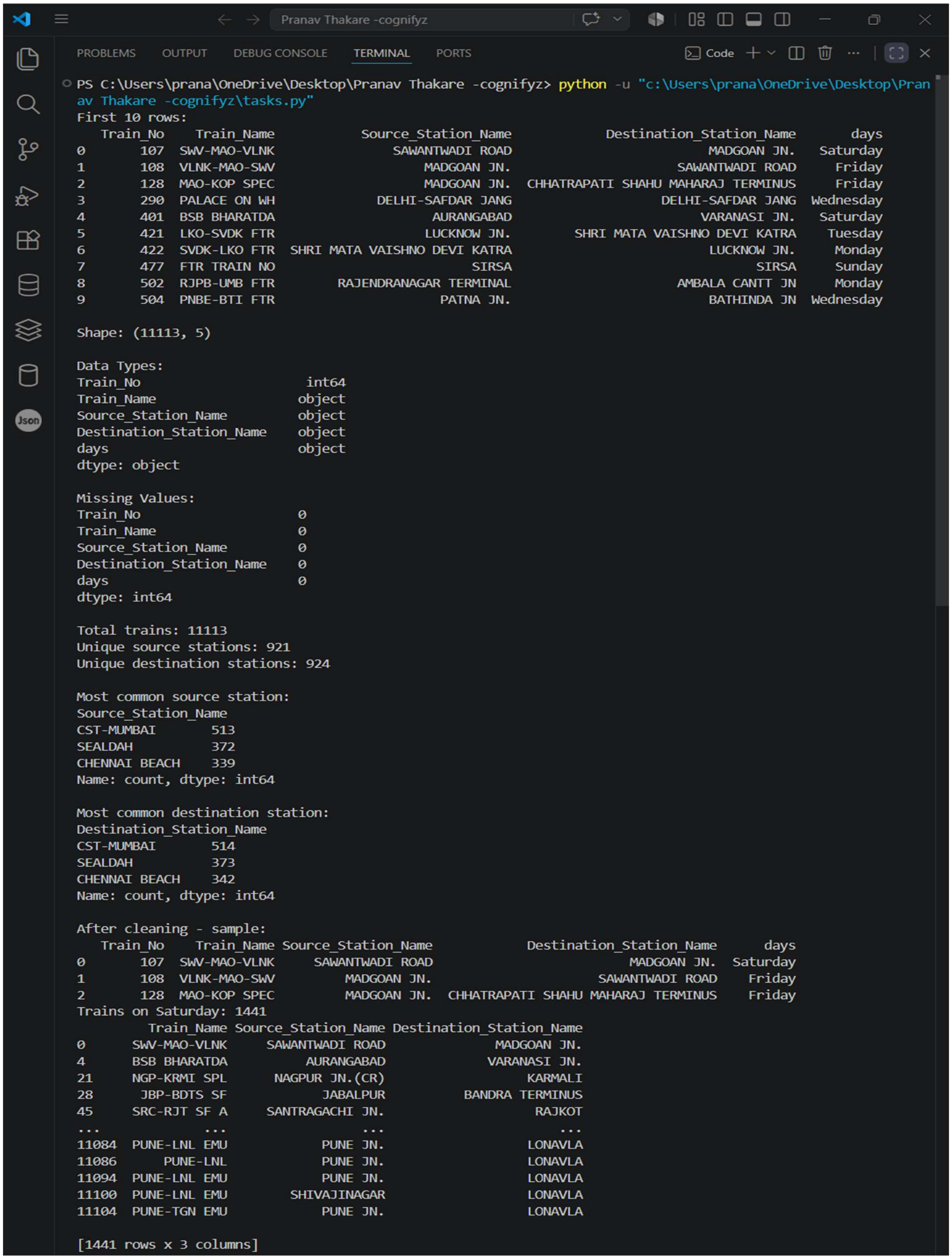
3. VISUALIZATIONS

Phase 1 Output:



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PS C:\Users\prana\OneDrive\Desktop\Pranav Thakare -cognifyz> python -u "c:\Users\prana\OneDrive\Desktop\Pran
av Thakare -cognifyz\tasks.py"
First 10 rows:
   Train_No     Train_Name          Source_Station_Name           Destination_Station_Name       days
0       107  SWV-MAO-VLNK               SAWANTWADI ROAD                      MADGOAN JN.   Saturday
1       108  VLNK-MAO-SWV                   MADGOAN JN.                  SAWANTWADI ROAD     Friday
2       128  MAO-KOP SPEC                   MADGOAN JN.  CHHATRAPATI SHAHU MAHARAJ TERMINUS     Friday
3       290  PALACE ON WH             DELHI-SAFDAR JANG                DELHI-SAFDAR JANG  Wednesday
4       401  BSB BHARATDA                    AURANGABAD                    VARANASI JN.   Saturday
5       421  LKO-SVDK FTR                   LUCKNOW JN.       SHRI MATA VAISHNO DEVI KATRA    Tuesday
6       422  SVDK-LKO FTR  SHRI MATA VAISHNO DEVI KATRA                     LUCKNOW JN.     Monday
7       477  FTR TRAIN NO                         SIRSA                            SIRSA     Sunday
8       502  RJPB-UMB FTR         RAJENDRANAGAR TERMINAL                  AMBALA CANTT JN     Monday
9       504  PNBE-BTI FTR                    PATNA JN.                      BATHINDA JN  Wednesday

Shape: (11113, 5)

Data Types:
Train_No                    int64
Train_Name                 object
Source_Station_Name        object
Destination_Station_Name   object
days                       object
dtype: object

Missing Values:
Train_No                   0
Train_Name                 0
Source_Station_Name        0
Destination_Station_Name   0
days                       0
dtype: int64

Total trains: 11113
Unique source stations: 921
Unique destination stations: 924

Most common source station:
Source_Station_Name
CST-MUMBAI        513
SEALDAH           372
CHENNAI BEACH     339
Name: count, dtype: int64

Most common destination station:
Destination_Station_Name
CST-MUMBAI        514
SEALDAH           373
CHENNAI BEACH     342
Name: count, dtype: int64

After cleaning - sample:
   Train_No     Train_Name Source_Station_Name           Destination_Station_Name      days
0       107  SWV-MAO-VLNK     SAWANTWADI ROAD                      MADGOAN JN.  Saturday
1       108  VLNK-MAO-SWV         MADGOAN JN.                  SAWANTWADI ROAD    Friday
2       128  MAO-KOP SPEC         MADGOAN JN.  CHHATRAPATI SHAHU MAHARAJ TERMINUS    Friday
Trains on Saturday: 1441
         Train_Name Source_Station_Name Destination_Station_Name
0      SWV-MAO-VLNK     SAWANTWADI ROAD              MADGOAN JN.
4      BSB BHARATDA          AURANGABAD             VARANASI JN.
21     NGP-KRMI SPL      NAGPUR JN.(CR)                  KARMALI
28      JBP-BDTS SF            JABALPUR          BANDRA TERMINUS
45     SRC-RJT SF A     SANTRAGACHI JN.                  RAJKOT
...             ...                 ...                      ...
11084  PUNE-LNL EMU           PUNE JN.                  LONAVLA
11086      PUNE-LNL           PUNE JN.                  LONAVLA
11094  PUNE-LNL EMU           PUNE JN.                  LONAVLA
11100  PUNE-LNL EMU        SHIVAJINAGAR                 LONAVLA
11104  PUNE-TGN EMU           PUNE JN.                  LONAVLA

[1441 rows x 3 columns]
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Phase 2 + 3 Output:

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PS C:\Users\prana\OneDrive\Desktop\Pranav Thakare -cognifyz> python -u "c:\Users\pran
a\OneDrive\Desktop\Pranav Thakare -cognifyz\tasks.py"
2335  NGP-BSP SHIV      NAGPUR JN.(CR)                                 BILASPUR JN.
2623  NGP-BSL INTE      NAGPUR JN.(CR)                                 BHUSAVAL JN.
2636  NGP-ASR AC E      NAGPUR JN.(CR)                                 AMRITSAR JN.
2644  NGP-REWA EXP      NAGPUR JN.(CR)                                         REWA
2646  PRERANA EXPR      NAGPUR JN.(CR)                                    AHMEDABAD
5664  NGP -WR PASS      NAGPUR JN.(CR)                                   WARDHA JN.
5668  NGP BSL PASS      NAGPUR JN.(CR)                                 BHUSAVAL JN.
5669  NGP AMLA PAS      NAGPUR JN.(CR)                                     AMLA JN.
5772   NGP-ET PASS      NAGPUR JN.(CR)                                       ITARSI
7524  NGP-RTK LOCA      NAGPUR JN.(CR)                                       RAMTEK

Top 10 source stations:
    Source_Station_Name  Train_Count
224           CST-MUMBAI          513
773              SEALDAH          372
201        CHENNAI BEACH          339
373           HOWRAH JN.          338
426            KALYAN JN          285
843                THANE          186
660               PANVEL          141
832             TAMBARAM          140
585          MOOR MARKET          135
894           VELACHEERY          115

Trains per day:
        days  Train_Count
0     Friday         1649
1     Monday         1503
2   Saturday         1593
3     Sunday         1602
4   Thursday         1526
5    Tuesday         1628
6  Wednesday         1612
Unknown days: []

Day category counts:
Day_Category
Weekday    7918
Weekend    3195
Name: count, dtype: int64
✅ Chart 1 saved
✅ Chart 2 saved
✅ Chart 3 saved

=== PHASE 3: PATTERN ANALYSIS ===

Train count by day:
days
Monday       1503
Tuesday      1628
Wednesday    1612
Thursday     1526
Friday       1649
Saturday     1593
Sunday       1602
Name: count, dtype: int64
✅ Histogram saved

Top 10 stations by avg trains per day:
    Source_Station_Name  Avg_Trains_Per_Day
224           CST-MUMBAI           73.285714
773              SEALDAH           53.142857
201        CHENNAI BEACH           48.428571
373           HOWRAH JN.           48.285714
426            KALYAN JN           40.714286
843                THANE           26.571429
660               PANVEL           20.142857
832             TAMBARAM           20.000000
585          MOOR MARKET           19.285714
894           VELACHEERY           16.428571

=== CORRELATION ANALYSIS ===
Correlation between day number and train count: 0.381
→ Moderate correlation found
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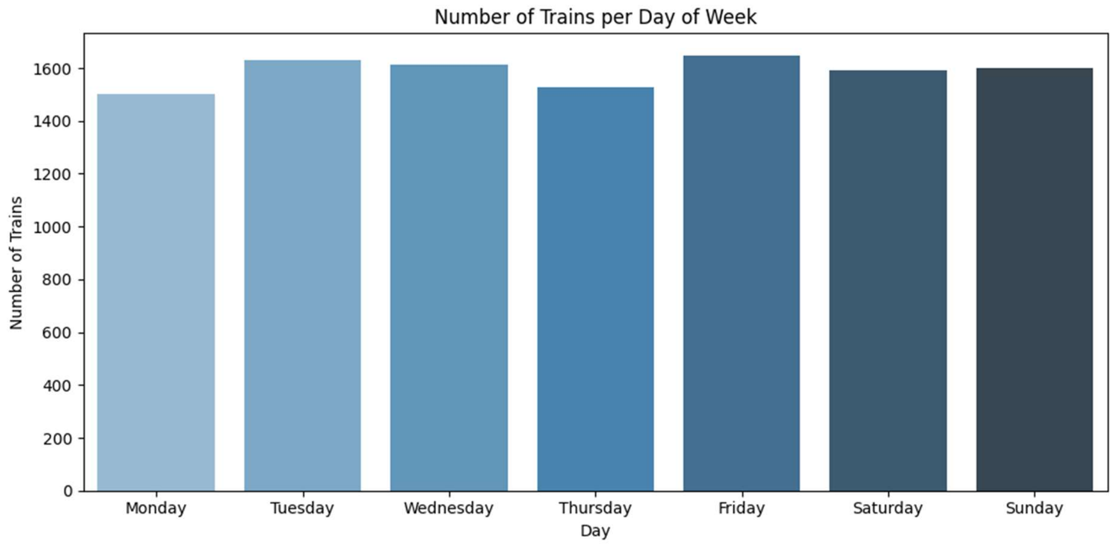
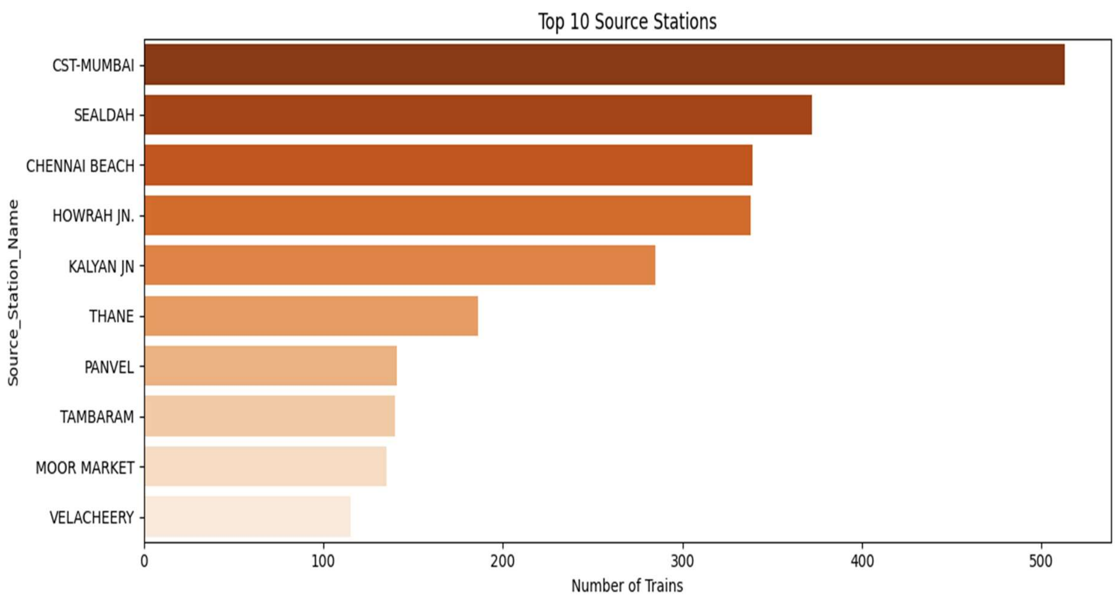
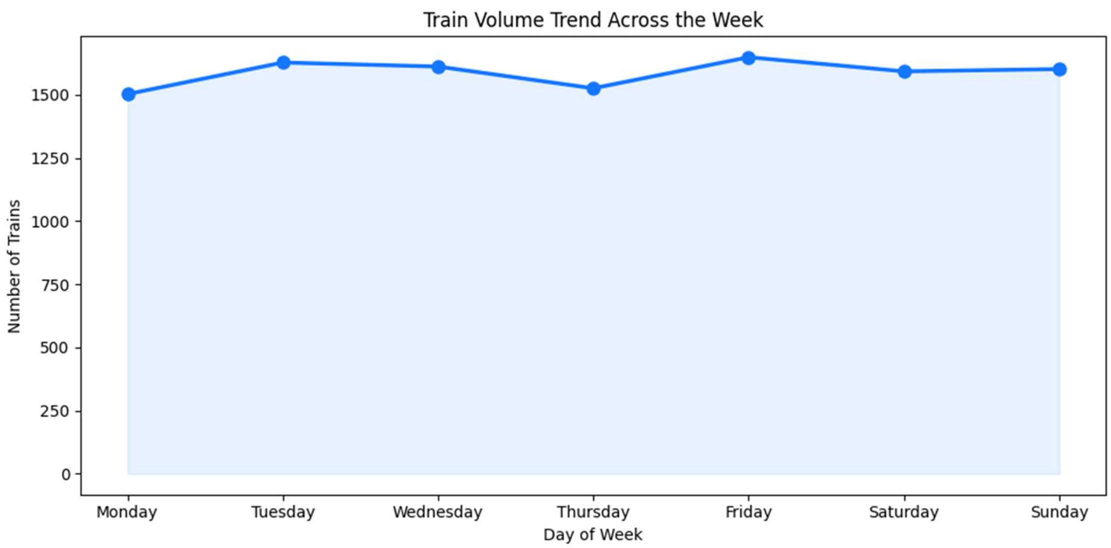
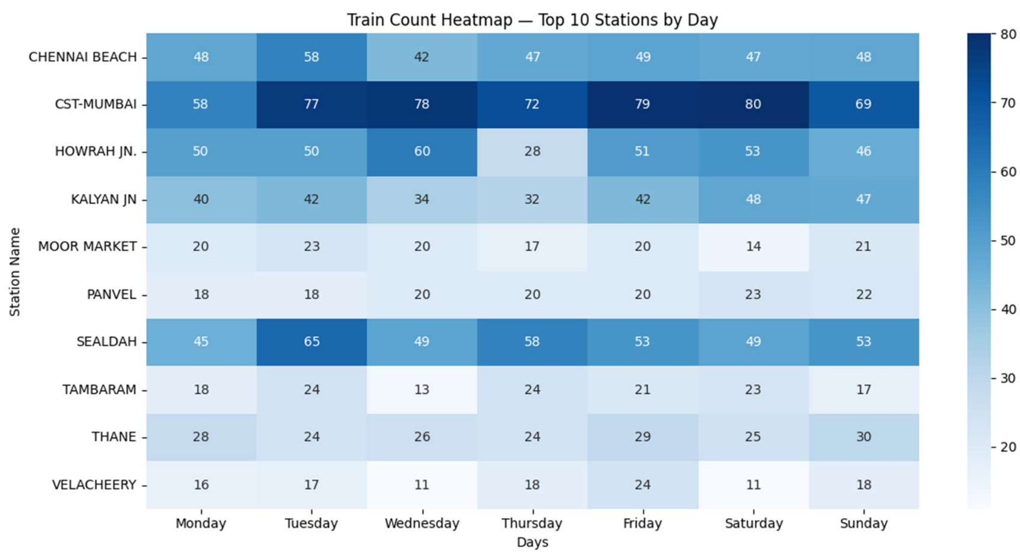
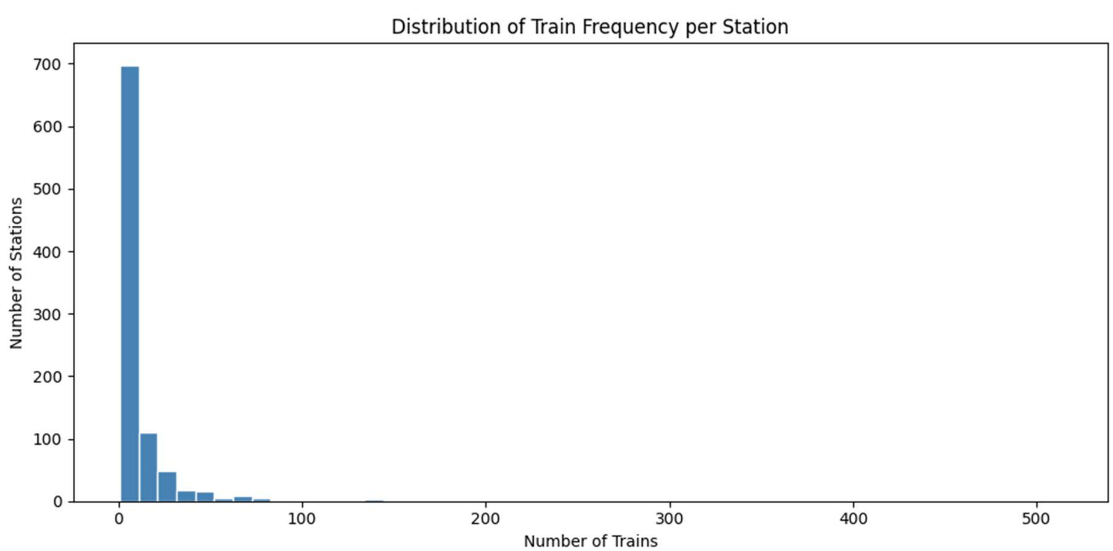
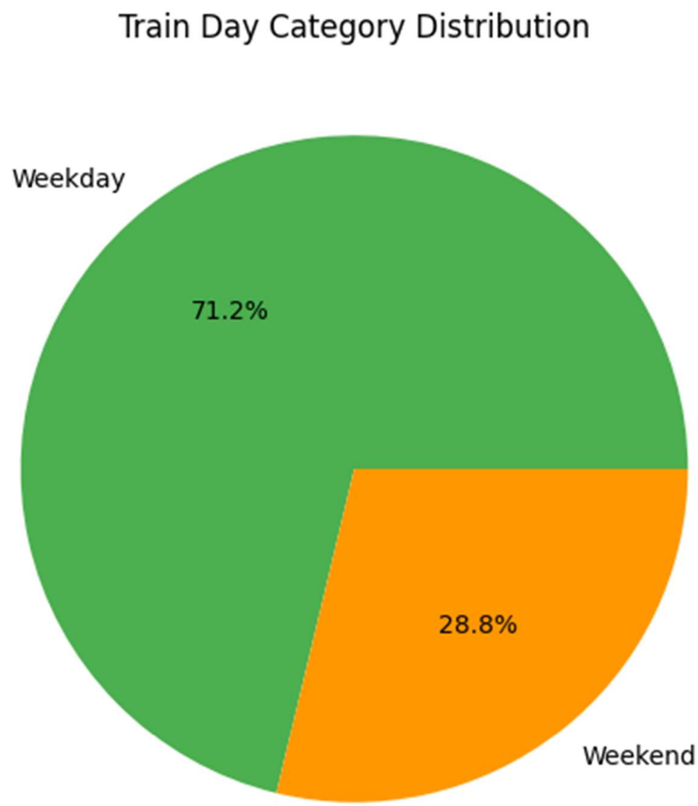

Number of Trains per Day of Week
Number of Trains
0
200
400
600
800
1000
1200
1400
1600
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Day


Top 10 Source Stations
Source_Station_Name
CST-MUMBAI
SEALDAH
CHENNAI BEACH
HOWRAH JN.
KALYAN JN
THANE
PANVEL
TAMBARAM
MOOR MARKET
VELACHEERY
0
100
200
300
400
500
Number of Trains


Train Volume Trend Across the Week
Number of Trains
0
250
500
750
1000
1250
1500
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Day of Week


Train Count Heatmap — Top 10 Stations by Day
Station Name
CHENNAI BEACH
CST-MUMBAI
HOWRAH JN.
KALYAN JN
MOOR MARKET
PANVEL
SEALDAH
TAMBARAM
THANE
VELACHEERY
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Days
48 58 42 47 49 47 48
58 77 78 72 79 80 69
50 50 60 28 51 53 46
40 42 34 32 42 48 47
20 23 20 17 20 14 21
18 18 20 20 20 23 22
45 65 49 58 53 49 53
18 24 13 24 21 23 17
28 24 26 24 29 25 30
16 17 11 18 24 11 18
80
70
60
50
40
30
20


Distribution of Train Frequency per Station
Number of Stations
0
100
200
300
400
500
600
700
0
100
200
300
400
500
Number of Trains


Train Day Category Distribution
Weekday
71.2%
28.8%
Weekend

4. RECOMMENDATIONS

1. Increase weekend trains to balance
   the 71% vs 29% weekday/weekend gap
2. CST-Mumbai needs infrastructure
   upgrades to handle high volume
3. Smaller stations (under 10 trains)
   need better connectivity

4. Thursday being lowest can be used
   for maintenance scheduling
5. Nagpur being a central location
   should have more connections added

5. CONCLUSION

The railway dataset reveals strong concentration of trains at major metro stations. Weekday operations dominate with moderate weekly variation. Strategic investment in weekend services and tier-2 city connectivity would significantly improve the overall railway network efficiency.